

UEC734: COMPILER CONSTRUCTION

L	T	P	Cr
3	0	0	3.0

Course Objectives: To Gain the working knowledge of the major phases of compilation and develop the ability to use formal attributed grammars for specifying the syntax and semantics of programming languages. Learn about function and complexities of modern compilers and design a significant portion of a compiler.

Introduction to Compiling

Analysis of the source program, Phases of a compiler, Cousins of the Compiler, Grouping of Phases, Compiler construction tools.

Lexical Analysis

Regular Expression, Introduction to Finite Automata and Regular Expression, Conversion of Regular Expression to NFA, Role of Lexical Analyzer, Input Buffering, Specification of Tokens.

Syntax Analysis

Role of the Parser, Writing Grammars, Symbol Table, Context-Free Grammars, Top Down Parsing with or without Backtracking, Recursive Descent Parsing, Non-Recursive Descent Parsing, SLR Parser, Canonical LR Parser, LALR Parser. Introduction to YACC and Integration with LEX.

Semantics Analysis and Intermediate Code Generation

Need for various static semantic analyses in declaration processing, name and scope analysis, Semantic analysis through S-attribute and L-attribute grammar, Intermediate languages, Declarations, Assignment Statements, Boolean Expressions, Case Statements, DAG representation of Basic Blocks, A simple Code generator from DAG, Issues in the design of code generator, The target machine, Runtime Storage management.

Error Handling Type checking, reporting and recovery from compilation errors, Classification of errors.

Code Optimization and Run Time Environments

Principal Sources of Optimization, Optimization of Basic Blocks, Peephole Optimization, Introduction to Global Data Flow Analysis, Source Language issues, Storage Organization, Static Storage Management, Heap Storage management, Access to non-Local Names, Parameter Passing.

Course Learning Outcomes (CLOs):

After the completion of the course, the student will be able to:

1. Comprehend the working of major phases of compiler.
2. Apply top-down and bottom-up parsing techniques for the Parser construction.
3. Understand the basic data structures used in compiler construction such as abstract syntax trees, symbol tables and three-address code
4. Understand target machine's run time environment and techniques used for code generation.

Text Books:

1. Aho V. A., Ullman D. J., Sethi R. and Lam S. M., Compilers Principles, Techniques and Tools, Pearson Education (2007), 2nd ed.
2. Levine J., Mason T., Brown D., Lex and Yacc, O'Reilly (2012), 2nd ed

UNC***: CYBER PHYSICAL SYSTEMS

L	T	P	Cr
2	0	2	3.0

Course Objectives: To develop the student's ability to understand the concept of cyber physical systems' characteristics, requirements and architecture.

Fundamentals of Cyber Physical Systems: Introduction to Cyber-Physical Systems (CPS) and Embedded systems, Industry 4.0 AutoSAR, IIOT implications, Building Automation, Agriculture CPS

Platform Components for Cyber Physical Systems: CPS HW platforms - Processors, Sensors, Actuators Si62 , CPS Network - WirelessHart, CAN, Automotive Ethernet, Scheduling-Real Time CPS tasks: Table-driven and Event driven schedulers Hybrid schedulers, Real time sensing and communication for CPS

Dynamical system modeling, stability, controller design: Dynamical Systems and Stability, Controller Design Techniques, Reachability analysis, Overview of Performance under Packet drop and Noise, CPS Performance Analysis: Effect of scheduling, bus latency, sense and actuation faults on control performance, network congestion Building real-time networks for CPS

Intelligent CPS: Overview of Safe Reinforcement Learning: Robot motion control, Autonomous Vehicle control, Lyapunov Stability, Barrier Functions, Neural Network (NN) Based controllers in CPS, State Estimation using Kalman Filters (KF)

Laboratory work: Hands-on on various sensors and actuators

Course Learning Outcomes (CLO):

The students will be able to:

1. Understand the key concepts, components, and applications of Cyber-Physical Systems (CPS)
2. Describe the hardware and network platforms used in CPS, including sensors, actuators, processors, and communication protocols.
3. Analyze real-time scheduling mechanisms and evaluate their implications on CPS task performance and reliability.
4. Model and assess the dynamics of cyber-physical systems using system stability, controller design techniques.
5. Design and implement intelligent CPS control strategies.

Text Books:

1. Suh, Sang C., U. John Tanik, John N. Carbone, and Abdullah Eroglu, eds. Applied cyber-physical systems. Springer New York, 2014.
2. Alur, Rajeev. Principles of cyber-physical systems. MIT Press, 2015.
3. Colombo, Armando W., Thomas Bangemann, Statmatis Karnouskos, Jerker Delsing, Petr Stluka, Robert Harrison, Francois Jammes, and Jose L. Lastra. "Industrial cloud-based cyber-physical systems." The Imc-aesop Approach 22 (2014): 4-5.